CIS 4190/5190: Applied Machine Learning Fall 2026

Homework 3

Handed Out: Tue October 20, 8:00 p.m. ET *Due: Tue November 10, 7:59 p.m. ET*

- Submit written HW3 as one legible PDF through Gradescope. You may write your answers in LaTeX or scan handwritten work. We cannot award credit for work that cannot be read.
- Assign the correct PDF pages to every question and sub-question in Gradescope. Unassigned pages may receive no credit.
- The submission deadline is **7:59 p.m. ET**. The late penalty is 0.5% of the assignment's maximum score per late hour (rounded up), for at most 48 late hours. Written and coding submissions are timed separately.
- The written and coding halves are graded separately. Written HW3 contains 55 points for CIS 5190 students. The 5-point part marked **[5190 Only]** is optional for CIS 4190 students, whose written maximum is therefore 50 points. Optional work cannot raise a score above 100%.
- Follow the Human and AI Assistance in Homework policy. Do not share or discuss solutions with other students. AI use is permitted only after a documented solo attempt and must be disclosed below.
- If using LaTeX, copy the shared solutions template. The link is read-only; make a *private* copy before editing.
- Post clarification questions on Ed Discussion. Contact the course staff promptly if you experience submission problems.

1 Use of AI Tools

Report any use of AI tools (ChatGPT, Claude, Copilot, Gemini, Cursor, etc.) while working on **this written homework**. You report AI use for the coding homework separately, in the reporting cell at the end of the notebook.

Being honest here will not cost you points. This section is graded for being answered, not for any particular answer: reporting that you used no AI earns full marks, and so does reporting that you used a lot of it. We are trying to learn what actually helps people learn, which only works if the reports are accurate.

1. (2 pts) Did you use AI tools on this written homework? If **no**, say so and you are done with this section. If **yes**, also answer:
   - Which tools did you use?

- **How** did you use them? Be specific: checking an answer you had already derived, explaining a concept you were stuck on, generating a first draft you then rewrote, doing the derivation for you, etc.
- How do you think this affected your learning of this material? Say so if you think it helped, and equally if you think it let you skip understanding something you would otherwise have worked out.

2 Time Spent

1. (2 pts) Roughly how long did each part of this **written** homework take, in hours? Decimals are fine (0.5 = 30 minutes), and enter 0 for any question you skipped. Report a time for each of Questions 1–5 as well as for reading the assignment. Rough estimates are genuinely useful — please do not agonize over precision, but do fill this in **as you go** rather than reconstructing it at the end. Also graded for being answered, not for what you answer.

3 Written Questions

You do not need to show work for multiple-choice items unless a question says otherwise. For derivations and calculation questions, show enough work for the grader to follow your reasoning; an unsupported final answer may not receive full credit.

1. [k-means] (10 pts) Consider the k-means clustering algorithm:

   (a) [8 pts] Work through the k-means clustering algorithm for a dataset with four samples, using $K = 2$ and the L_2 distance. The samples are $A = (2, 3)$, $B = (4, 6)$, $C = (5, 1)$, and $D = (10, 12)$. The initial centroids are $(6, 9)$ for cluster 1 and $(8, 4)$ for cluster 2. Each iteration has two steps: first compute distances and update cluster assignments using the centroids at the start of that iteration, and then update the centroids. Use the initial centroids for iteration 1 and the centroids found at the end of iteration 1 for iteration 2. Report the following for each iteration:

   - A: $d(A, 1)$, $d(A, 2)$
   - B: $d(B, 1)$, $d(B, 2)$
   - C: $d(C, 1)$, $d(C, 2)$
   - D: $d(D, 1)$, $d(D, 2)$
   - cluster 1 members: A, B, etc.
   - cluster 1 updated centroid: (x, y)
   - cluster 2 members: A, B, etc.
   - cluster 2 updated centroid: (x, y)

   where $d(S, c)$ is the L_2 distance from sample S to the cluster c centroid. Give each distance as an exact radical; decimal approximations are optional.

(b) [2 pts] Consider what happens when $K = 4$. Assume we run k-means with an initialization that achieves the global minimum loss. What will be the coordinates of the four learned centroids? (Hint: you do not need to calculate anything like in part (a).)

2. [PCA] (10 pts) *Note: You are expected to work out the entire question by hand and not use any libraries or packages. Plots may be drawn by hand or with drawing tools, but you must determine the labels and lines yourself.*



Figure 1: Draw the first and second principal components on each plot.

(a) [5 pts] Principal component analysis is a dimensionality reduction method that projects a dataset onto directions of greatest variance. For each 2-D dataset in Figure 1, draw the first and second principal components and label them PC1 and PC2. *Note: The different colors are used only for visualization.*

(b) [5 pts] **[5190 Only]** Assume we are given a dataset for which the eigenvalues of the covariance matrix are: (2.1, 1.8, 1.3, 0.9, 0.4, 0.2, 0.15, 0.02, 0.001). What is the smallest value of K (the dimension after reduction) that retains at least 75% of the total variance when using the leading principal components? Justify your answer; you may use a four-function calculator.

3. [PCA] (13 pts) *Note: You are expected to work out the entire question by hand and not use any libraries or packages. Plots may be drawn by hand or with drawing tools, but you must determine the labels and lines yourself.*

Bob wants to transmit the following set of four two-dimensional coordinates to his friend.

$$X = \begin{bmatrix} 4 & 1 \\ 2 & 3 \\ 5 & 4 \\ 1 & 0 \end{bmatrix}$$

However, due to transmission bandwidth, he is constrained to send only four one-dimensional coordinates. He learns about the PCA algorithm and wants to apply it.

(a) [8 pts] Find the unit-vector principal components of X. Given that Bob is constrained to send four 1-D coordinates, which principal component would you suggest him to pick, and why? Show your work.

(b) [5 pts] Bob plots the coordinates of X as shown in Figure 2. In the plot, $A = (4, 1)$, $B = (5, 4)$, $C = (2, 3)$, and $D = (1, 0)$. First center the data by subtracting the mean of the four points. Compute each 1-D principal coordinate by projecting a centered point onto the principal-component direction from part (a). Sketch the projections on Bob's plot, drawing the principal-component line through the data mean, and label each projected point with its 1-D principal coordinate.

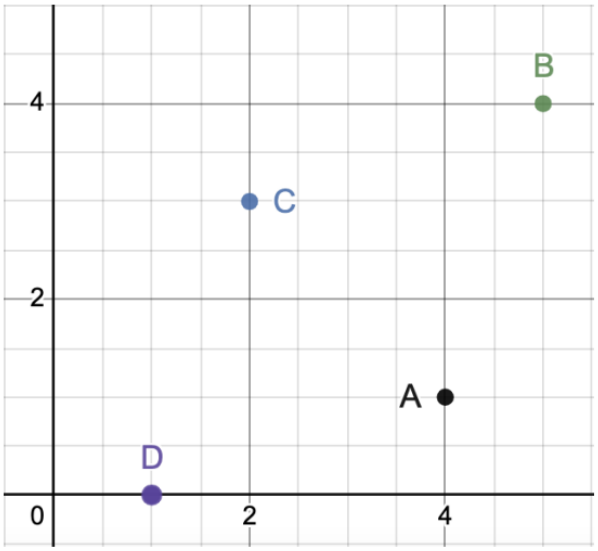


Figure 2: Bob's plot

4. [Image Filtering/Convolution] (9 pts) We briefly discussed image convolution filters in class, primarily in the context of CNNs. Before CNNs became popular, convolution filters were already essential tools in signal processing and computational photography. Many features in Photoshop or Lightroom can be implemented using an appropriate convolution filter. In this question, you will examine several common filters and visualize how they transform an image. The example below is a grayscale image. Each pixel has a value in $[0, 1]$, where 0 is black and 1 is white. The image can be represented by a matrix with shape height $\times$ width. You will apply 3×3 convolution filters to this matrix; assume that the bias is 0 for every filter.

   For each sub-question, choose the transformed image that corresponds to the given convolution filter.



(a) [3 pts] Consider the following filter:

$$X = \begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix}$$

If we apply X as a convolution filter to the original image, which of the following transformed images will we see? **Briefly justify your choice using one or two sentences.**

A. 

B. 

C. 

D. 

(b) [3 pts] Now consider the following filter. Use the pixel values in $[0, 1]$ defined above; for display, clip filter outputs below 0 to 0 and outputs above 1 to 1:

$$X = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

If we apply X as a convolution filter to the original image, which of the following transformed images will we see? **Briefly justify your choice using one or two sentences.**

A.  B. 

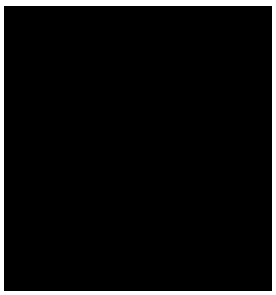
C.  D. 

(c) [3 pts] Now let's look at a more challenging example:

$$X = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$

If we apply X as a convolution filter to the original image, which of the following transformed images will we see? **Briefly justify your choice using one or two sentences.**

A.  B. 

C.  D. 

5. [CNNs] (9 pts) Consider the following neural network with three convolutional layers.

   i. **Input**: RGB image $320 \times 320 \times 3$
   ii. **Conv2d**: 5×5 kernel, Stride=1, Padding=2, Out channels=16
   iii. **ReLU**
   iv. **MaxPool2d**: kernel: 2x2, Stride 2, Padding 0
   v. **Conv2d**: Out channels: 8, kernel: 3x3, Stride 1, Padding 1
   vi. **ReLU**
   vii. **MaxPool2d**: kernel: 2x2, Stride 2, Padding 0
   viii. **Conv2d**: Out channels: 4, kernel: 3x3, Stride 1, Padding 1
   ix. **ReLU**
   x. **MaxPool2d**: kernel: 2x2, Stride 2, Padding 0

   For this question, each convolutional layer has one learnable bias per output channel. Include these biases when counting parameters in part (d).

   If the input image is 320×320 pixels with RGB channels (3 channels), compute the following:

   (a) [2 pts] Output height

   (b) [2 pts] Output width

   (c) [2 pts] Number of output channels

   (d) [3 pts] Number of learnable parameters